



Cartography M.Sc.



Seminar Paper

Communicating Uncertainty in Flood Hazard Maps for Non-Expert Users

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Abstract

Flood maps that show predicted flood extents will always be uncertain regardless of how the modelling is conducted. It is therefore important that these uncertainties are represented and communicated in the maps, and that map users understand the presented information. This paper examines how uncertainty in flood hazard maps can be communicated effectively to non-expert users without reducing map comprehension, usability, or decision relevance. It uses a structured review of selected literature on flood-risk communication, uncertainty visualisation, hazard-map comprehension, and user-centred cartographic design. The review shows that uncertainty should be communicated clearly, but only in forms that users can interpret and connect to decisions. Also, the most effective map design prioritise intuitive visual design, clear legend language, limited technical complexity, local reference points, and testing with the intended users. Blue-based and familiar colour schemes, plain descriptive categories, and integrated uncertainty displays are more suitable for non-expert users than technical probability labels, dense map layers, or separate uncertainty graphics. The paper concludes that uncertainty in flood maps should be communicated in a way that is both scientifically accurate and easy for the public to understand. Flood maps should not hide uncertainty. Instead, they should show it in simple, decision-relevant visual and written forms that help non-expert users understand the risk and prepare for possible flooding.

Keywords: flood hazard maps; non-expert users; flood-risk communication; map design; uncertainty; usability; map comprehension; decision relevance;

1. Introduction

The globally increasing frequency of flood events highlights the importance of effective flood risk communication. Flood hazard maps are among the most widely used spatial tools in flood-risk communication. By law, all EU Member States are required to produce and publish flood hazard maps. This makes maps both technical assessment products and public communication tools [3]. They support spatial planning, emergency planning, local hazard assessment, the design of technical protection measures, and public decision-making under uncertain flood conditions [12].

In many cases, these maps are intended for non-expert users rather than specialists alone. However, studies in flood-risk communication show that many official map products do not match the interpretive needs of their intended users because they fail to achieve a usable balance between simplicity and complexity [1]. Meyer et al. [3] similarly argued that flood maps are often treated primarily as information products rather than as communication tools tailored to end-user needs. Participatory studies also show that at-risk residents need clearer, more practical, and locally meaningful information than what many static map products currently provide [4], [7].

This raises an important concern: can non-expert users understand, trust, and act on the information presented in these maps? Flood maps are not direct pictures of future floods. They are estimates based on models, data, assumptions, and design choices. Nevertheless, many flood maps still hide uncertainty and use crisp boundaries and exact-looking zones that imply a level of precision the underlying science cannot fully justify.

Therefore, this paper asks: how can uncertainty in flood hazard maps be communicated effectively to non-expert users without reducing map comprehension, usability, and decision relevance?

2. Related Literature

2.1 Why Flood-Risk Communication Fails for Non-Expert Users

Risk communication involves the interactive exchange of information about the nature of risks, risk-related opinions, concerns, and coping strategies [1]. In a systematic review of 57 empirical studies, Kellens et al. [2] concluded that flood-risk management increasingly depends on subjective aspects of risk, including how people perceive hazards, seek information, and respond to warning signs. Their review also shows that theoretical and empirical research on flood-risk communication is scarce and methodologically varied, which makes comparison across studies difficult. This suggests that communication problems in flood management cannot be addressed through technical analysis alone.

Rollason et al. [4] similarly argue that existing flood-risk communication often fails to produce the intended improvements in awareness and behaviour because it does not provide the kinds of information that at-risk communities need. Their study suggests that users need to understand flood dynamics, including when flooding may occur, how it may unfold, and how they should respond, rather than only whether a location is inside or outside a risk zone. Yousaf et al. [12] also argue that flood-risk maps often fail because of poor readability, technical language, inconsistent colour conventions, and cognitive biases that affect how map readers interpret uncertainty and risk. Figure 1 illustrates how risk perception, map design, communication approach, and flood experience interact in a continuous cycle [12].

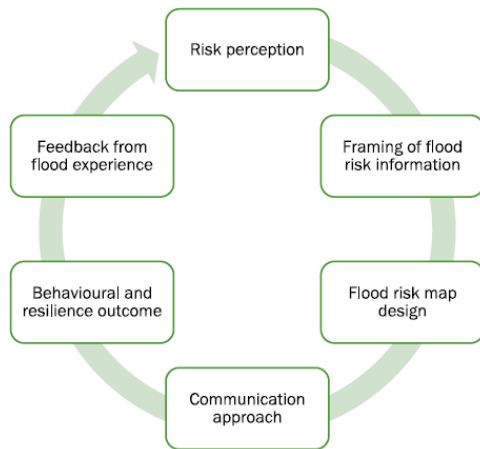


Figure 1 Conceptual flow diagram illustrating how risk perception theories shape flood risk map design and communication outcomes. Source: Yousaf et al. [12].

2.2 Uncertainty and Uncertainty Communication

Doyle et al. [5] defined uncertainty as an unavoidable feature of natural-hazard modelling. It may arise from natural variability, incomplete data, model structure, parameter choices, boundary conditions, and disagreement among model outputs. They identified several recurring themes in the communication of model uncertainty for natural hazards, including the need for clear uncertainty typologies, the need to involve users in deciding which uncertainties matter to them, and the limited evaluation of many current communication approaches. Their review shows that effective uncertainty communication depends not only on display format, but also on decision context and the relevance of the uncertainty for users. Kinkeldey et al. [14] also reviewed empirical user studies on geospatial uncertainty visualisation and concluded that assessment methods are often fragmented and difficult to compare. They advocate a more task-centred approach because no single uncertainty display is effective outside its intended use. This is important for flood maps designed for non-expert users, who are less likely to use maps for technical analysis and more likely to use them for judgements about location, exposure, preparedness, and everyday decision-making.

Although most literature treats uncertainty as a modelling and representation challenge, the interpretive issue is broad. Public flood maps may also be uncertain about future climatic conditions, changing land use and exposure, the effectiveness of protective infrastructure, and other human interventions. Cartographic design therefore needs to make uncertainty and contingent boundaries understandable to non-expert users, because these distinctions are often not obvious on maps [5], [11].

In this paper, non-expert users are defined as members of the public, residents in flood-prone areas, and other map readers without specialist training in hydrology, GIS, or cartography. Map comprehension is defined as how accurately users read and interpret hazard information from a map [6]. Usability refers to how easily and confidently users can process information from map and act on it [9]. Decision relevance refers to whether the map information supports the specific choices users need to make [5], [8].

3. Review Approach

This paper uses a structured literature review as its methodological approach. The review synthesises selected research papers to answer the research question, rather than attempting to cover the entire field of flood-risk and uncertainty communication. Priority was given to peer-reviewed papers that addressed at least one of the following criteria: (1) flood hazard or flood-risk maps, (2) uncertainty representation

and communication, (3) non-expert comprehension, usability, or decision-making, and (4) practical cartographic design for non-expert users. Table 1 summarises the methodological profile of the main studies used in the analysis. The reviewed literature includes systematic and narrative reviews, participatory studies, resident surveys, controlled experiments, and user-centred design research.

Table 1 Methodological profile of the reviewed studies

Study	Type	Study context	Main relevance
Hagemeier-Klose and Wagner [1]	Mixed-method evaluation	Existing tools analysis; workshop with experts and lay users; online survey	Public readability, usability, and intuitive framing
Kellens et al. [2]	Systematic review	57 empirical flood-risk studies	Flood-risk perception and communication background
Meyer et al. [3]	Participatory mixed methods	Interviews, workshops, and eye-tracking in five European case studies	User-specific flood-map enhancement
Rollason et al. [4]	Participatory experiment	Local competency group in an at-risk UK community	User needs, flood dynamics, and response relevance
Doyle et al. [5]	Qualitative systematic thematic review	Cross-hazard literature	Decision-relevant uncertainty communication
Lindell [6]	Narrative review	Hazard-map comprehension studies	Inference errors and protective decision-making
Percival et al. [7]	Co-creative workshops	Stakeholder workshops over 12 months	Effective flood-risk visualisation criteria
Twomlow et al. [8]	Systematic review and framework	Disaster-risk visualisation literature	User-centred define-design-refine framework
Stephens et al. [9]	Resident survey	N = 404 Texas residents in flood-affected counties	Usability, legends, and perceived accuracy
Kuster et al. [10]	Between-subject experiment	Public users; 2D versus 2.5D flood maps	Viewing perspective and risk perception
Lim et al. [11]	Online user survey	Users evaluated nine flood uncertainty maps	Representation effects on decisions and confidence
Yousaf et al. [12]	Narrative/scoping review	Conceptual and empirical flood-risk literature	Challenges, framing, trust, and participatory mapping
Mulder et al. [13]	Decision experiment	N = 138 experts and non-experts	Format effects on inferred uncertainty
Kinkeldey et al. [14]	Systematic review	Geospatial uncertainty-visualisation user studies	Task-centred evaluation concepts

4. Findings and Discussion

4.1 Uncertainties Relevant to Decision-making Must Be Communicated in Flood Hazard Maps

One consistent finding across the reviewed literature is that uncertainty should be communicated explicitly rather than hidden. Lim et al. [11] showed that users' choices, confidence levels, and task performance vary with the design of uncertainty maps. This means that uncertainty representation is not a cosmetic layer, but part of the decision environment itself. Doyle et al. [5] reached a similar theoretical conclusion, arguing that failing to communicate decision-relevant uncertainty can mislead users about the limits of natural-hazard evidence. Focusing specifically on flood-risk maps, Yousaf et al. extended this argument by noting that probabilistic maps can better represent the spatial variability of flood risk, if users can understand what the probabilities mean [12].

For non-expert users, this means all uncertainty cannot be displayed at once. The most useful flood hazard maps will highlight key uncertainties for public interpretation, such as property proximity to flood boundaries, approximate depth or probability, and reliance on rainfall, flood defenses, maintenance, land-use conditions, and return-period assumptions. Communicating these uncertainties clearly helps users understand that the map is only a guide to possible conditions, not a precise prediction of a future event. In this regard, effective uncertainty communication is selective but tied to the decisions non-experts need to make, such as recognizing possible exposure, comparing relative seriousness, and judging whether further information or protective action is needed. Figure 2 shows how a binary map communicates only whether an area is at risk, while a probabilistic map shows graduated risk levels with annual probabilities, giving users more decision-relevant information [12].

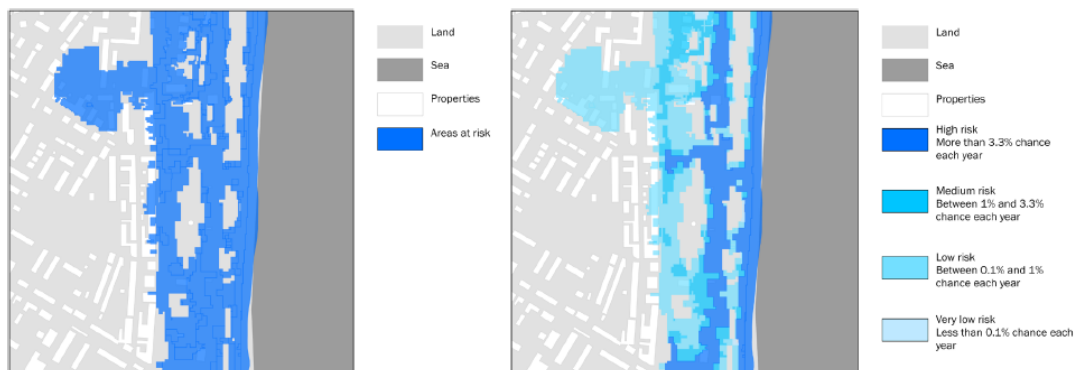


Figure 2 Binary flood map versus probabilistic flood map of the same area. Source: Yousaf et al. [12].

4.2 Cartographic Designs That Support Uncertainty Interpretation.

Showing uncertainty on a map is only useful when users can understand what it means. Flood-specific studies consistently showed that readability, familiar language, and local meaning are key to effective public maps [1], [3]. Hagemeyer-Klose and Wagner found that users understand flood information better when visual categories are intuitive and connected to familiar reference points such as gauge levels or past events [1]. Stephens et al. [9] also found that perceived usability strongly predicts perceived map accuracy, indicating that trust depends partly on whether the map can be processed easily. Percival et al. similarly emphasized consistent terminology and centralized, relevant information as conditions for effective flood-risk visualization [7].

These findings suggest an important design balance. A flood hazard map should not be so technical that it overwhelms non-expert users, nor should it be so simple that it removes the context users need for decisions. Colour and wording are not minor styling choices. They are part of how the map explains risk. Intuitive water-related colors, clear contrast between classes, and short legends can support easy interpretation, while poorly chosen gradients or dense statistical terms can reduce confidence and increase error. For example, a legend labeled as 'possible flooding in extreme rainfall' is easier for a resident to use than a purely statistical one. The map should tell users not only that flooding is uncertain, but why that uncertainty matters for their location, route, property, or preparedness decision. Taken together, these design principles directly address the research question by showing that uncertainty can be made interpretable without removing the information users need to act.

Figure 3 shows three legend formats tested by Stephens et al. [9], where the Flood Hazard Index was rated most useful by participants.

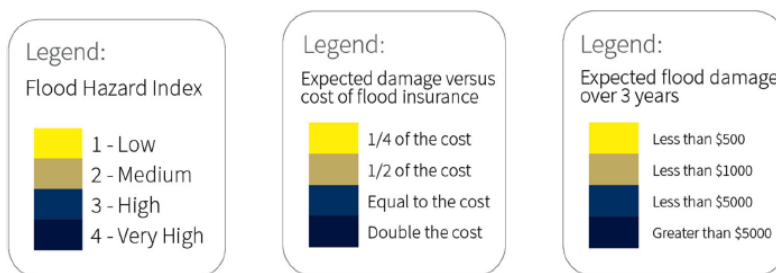


Figure 3 Three flood map legend formats tested with residents. The Flood Hazard Index was rated most useful. Source: Stephens et al. [9].

4.3 Design-Centred and Participatory Testing

Meyer et al. showed that different audiences need different flood-map products, because planners, emergency managers, and residents do not ask the same questions and have different needs [3]. Rollason et al. and Percival et al. demonstrated that participatory studies could reveal what static maps often miss: residents want to understand when and how flooding might happen, what it could mean for their area, and what action they should take in response [4], [7]. Twomlow et al. synthesized these lessons in a Define–Design–Refine framework that begins with audience and context, iterates through testing, and ends with refinement based on whether users understand and act on the visualization [8].

4.4 Answering the Research Question Through Map Design and Testing

The research question can be answered as: uncertainty should be communicated to non-expert users in simple, task-oriented, and action-relevant visual and verbal forms. A public-facing map should show that flood boundaries and probabilities are uncertain. It should do so with plain-language legends, intuitive colour choices, local reference points, and short explanations of the scenario. More detailed uncertainty information can be placed in supporting panels, while the main map should remain simple for the user who needs to decide whether they are exposed, how serious the hazard may be, and what action to consider.

A layered map arrangement follows from this. The first layer should provide the clearest public message: where flooding may occur and how severe it may be. The second layer should explain the uncertainty behind that message, for example by showing a boundary band, probability classes, or a short statement about modelling assumptions. A third optional layer can provide technical detail for users who want it, such as model scenario, return period, data date, and links to emergency guidance. This layered structure keeps the main map readable while still making uncertainty transparent.

The same principle applies to format choice. Static printed maps may need very compact uncertainty explanations, while web maps can use pop-up notes, toggled uncertainty layers, and examples tied to known places. In both cases, the communication goal should help a non-expert see that risk changes gradually across space and that the mapped boundary is not an exact line between safety and danger.

The review also shows that a map cannot be assumed to communicate uncertainty well without testing it with real users. The reviewed studies used different methods, including reviews, workshops, surveys, and experiments, so the findings cannot be treated as a universal formula. However, they all agree on one key principle: scientific honesty and public usability are not opposing goals. Both can be achieved if map producers focus on the uncertainty that matters most, express it in a form non-experts understand, and verify through testing that the map is read as intended [6], [8], [9], [11].

The answer is a design process and not a single symbol system. Map producers should begin by defining the decision the public must make, choose only the uncertainty information relevant to that decision, express it through familiar visual and verbal conventions, and then revise the map after observing how intended users interpret it. Therefore, the reviewed literature gives a consistent answer for cartographic practice: uncertainty should be visible, explained in ordinary language, connected to local decisions, and refined through user testing. These conditions allow flood maps to remain scientifically honest while still being usable communication tools for people without technical training in GIS, hydrology and Cartography.

5. Main Message

The key message from the reviewed literature is that agencies should not begin with the question, "How much uncertainty can we display?" Instead, they should ask, "What information about uncertainty does the user need in order to understand the risk and respond appropriately?" Table 2 summarises the main design principles and their relevance for public-facing flood hazard maps.

Table 2 Key design rules of Flood hazard map for communicating uncertainty for non-expert users

Design rule	Why it matters
Start with users and their needs	Define who will use the map, what decision they need to make, and the time pressure under which the map will be used.
Show uncertainty explicitly	Avoid implying that flood boundaries are exact but show uncertainty in a form users can interpret.
Use intuitive color semantics	Use colors with familiar associations to water and risk, and test whether users interpret the map as intended.
Use clear legend language	Prioritize descriptive categories and short explanations over specialist statistical labels where possible.
Keep maps simple but decision-relevant	Avoid unnecessary technical clutter while retaining the contextual information needed for action.
Provide local reference points	Connect map classes to known places, landmarks, gauge levels, or past events where appropriate.
Do not rely on a single static map	Embed the map in a broader communication strategy that includes explanation, alerts, and supporting channels.
Test with non-experts before release	Evaluate comprehension, trust, and intended action among target users rather than assuming expert readability.

6. Limitations and Future Research

This paper has several limitations. It is a structured literature review of selected papers rather than a complete systematic review; therefore, depth of comparison is prioritised over completeness. The reviewed studies also differ in methods, geographical settings, user groups, and outcome measures, which limit how far their findings can be generalized. Stephens et al. [9] surveyed residents in Texas flood-affected counties, which limits how far their findings on legend usability can be generalised to other cultural or geographic settings. Lim et al. [11] used an online survey to test uncertainty map formats, which may not fully reflect how users behave under real flood conditions. Rollason et al. [4] worked with a single local community group in the United Kingdom, making their participatory findings context specific. These limitations do not invalidate the findings, but they do mean that the design principles identified in this paper should be treated as evidence-informed guidance rather than universal rules.

Future research should focus on these areas. First, more empirical studies are needed on how the public interprets specific uncertainty map types during decision making. Second, effective communication of uncertainties, such as climate-change-driven shifts in event frequency, changes in land use, and the changing reliability of structural protection in many flood hazard maps intended for non-expert users to support decision making. Third, studies should clearly describe their participants, so we know whether they truly represent non-expert users.

7. Conclusion

Flood maps that show predicted flood extents will always be uncertain regardless of how the modelling is conducted. It is therefore important that these uncertainties are represented and communicated in the maps, and that map users understand the presented information. Based on the reviewed literature, three main conclusions can be drawn. First, uncertainty should be communicated explicitly, but only in forms that are interpretable and decision-relevant for non-expert users. Second, usability is central to communication effectiveness because users are more likely to trust and act on maps they can easily use. Third, uncertainty communication should follow a user-centred approach that combines intuitive visual distinctions, accessible legend language, local reference points, and iterative testing with intended users. Hence, these three (3) conclusions answer the research question: *how can uncertainty in flood hazard maps be communicated effectively to non-expert users without reducing map comprehension, usability, and decision relevance*

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